

2023—2024第二学期《高等数学》期中试题参考解答

1. 解: 由已知条件可得 $\vec{m} \cdot \vec{c} = (\lambda \vec{a} + \vec{b}) \cdot \vec{c} = \lambda \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} = 0$,

$$\Rightarrow \lambda = -\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{c}} = \frac{5}{4}.$$

2. 解: 由向量混合积的几何意义知以 $\vec{AB}, \vec{AC}, \vec{AD}$ 为棱的平行六面体的体积为 $|\vec{AB}, \vec{AC}, \vec{AD}|$,

故所求四面体的体积为

$$V = \frac{1}{6} |\vec{AB}, \vec{AC}, \vec{AD}| = \frac{1}{6} \begin{vmatrix} -1 & 1 & 3 \\ -3 & 4 & 6 \\ 1 & -2 & 4 \end{vmatrix} = \frac{2}{3}.$$

3. 解: (1). 因为 $0 \leq |f(x, y)| = \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| \leq |y|$, 且 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} |y| = 0$,

所以 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = 0 = f(0, 0)$, 即 $f(x, y)$ 在点 $(0, 0)$ 连续.

$$(2). f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0}{x} = 0,$$

$$f'_y(0, 0) = \lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0}{y} = 0,$$

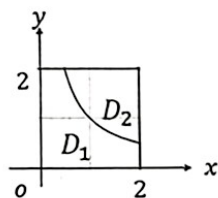
即 $f(x, y)$ 在点 $(0, 0)$ 处可偏导, 且偏导数为 0.

$$(3). \because \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x, y) - f(0, 0) - f'_x(0, 0)x - f'_y(0, 0)y}{\rho} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2 + y^2} \neq 0,$$

$\therefore f(x, y)$ 在点 $(0, 0)$ 不可微.

4. 解: 如图所示, 在积分区域 D 内添加辅助线 $xy = 1$ 将积分区域分为两部分 D_1 和 D_2 , 则

$$\begin{aligned} \iint_D \max\{xy, 1\} dx dy &= \iint_{D_1} dx dy + \iint_{D_2} xy dx dy \\ &= \iint_D dx dy + \iint_{D_2} (xy - 1) dx dy \\ &= 4 + \int_{\frac{1}{2}}^2 dx \int_{\frac{1}{x}}^2 (xy - 1) dy = \frac{19}{4} + \ln 2. \end{aligned}$$



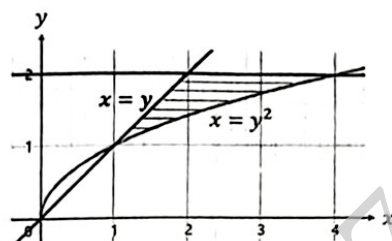
5. 解: 在积分区域 D 内添加辅助线 $x^2 + y^2 = 4$, 将 D 分为 D_1 和 D_2 , 其中

$D_1 = \{(x, y) | x^2 + y^2 \leq 4\}$, $D_2 = \{(x, y) | 4 < x^2 + y^2 \leq 9\}$, 则

$$\begin{aligned} \iint_D |x^2 + y^2 - 4| dx dy &= \iint_{D_1} (4 - x^2 - y^2) dx dy + \iint_{D_2} (x^2 + y^2 - 4) dx dy \\ &= \int_0^{2\pi} d\theta \int_0^2 (4 - \rho^2) \rho d\rho + \int_0^{2\pi} d\theta \int_2^3 (\rho^2 - 4) \rho d\rho = \frac{41}{2} \pi. \end{aligned}$$

6. 解: 积分区域如图所示, 交换积分次序得

$$\begin{aligned} \text{原式} &= \int_1^2 dy \int_y^{y^2} \sin \frac{\pi x}{2y} dx = \frac{2}{\pi} \int_1^2 y dy \int_y^{y^2} \sin \frac{\pi x}{2y} d\left(\frac{\pi x}{2y}\right) \\ &= -\frac{2}{\pi} \int_1^2 y \left[\cos \frac{\pi x}{2y} \right]_y^{y^2} dy = -\frac{2}{\pi} \int_1^2 y \cos \frac{\pi y}{2} dy \\ &= -\frac{4}{\pi^2} \int_1^2 y d\left(\sin \frac{\pi y}{2}\right) = -\frac{4}{\pi^2} \left(y \sin \frac{\pi y}{2} \Big|_1^2 - \int_1^2 \sin \frac{\pi y}{2} dy \right) \\ &= \frac{4}{\pi^2} + \frac{8}{\pi^3}. \end{aligned}$$



7. 解: 联立 $\begin{cases} z = \sqrt{x^2 + y^2} \\ z = \sqrt{1 - x^2 - y^2} \end{cases}$ 得 $\begin{cases} x^2 + y^2 = \frac{1}{2} \\ z = \frac{\sqrt{2}}{2} \end{cases}$, 则

$$\iiint_{\Omega} \sqrt{(x^2 + y^2)} dx dy dz = \int_0^{2\pi} d\theta \int_0^{\frac{\sqrt{2}}{2}} \rho d\rho \int_{\rho}^{\sqrt{1-\rho^2}} \rho dz = \frac{\pi}{16} (\pi - 2).$$

或者:

$$\iiint_{\Omega} \sqrt{(x^2 + y^2)} dx dy dz = \int_0^{\frac{\sqrt{2}}{2}} dz \int_0^{2\pi} d\theta \int_0^z \rho^2 d\rho + \int_{\frac{\sqrt{2}}{2}}^1 dz \int_0^{2\pi} d\theta \int_0^{\sqrt{1-z^2}} \rho^2 d\rho = \frac{\pi}{16} (\pi - 2).$$

8. 解: 设长方体位于第一卦限的顶点为 (x, y, z) , 则其体积为 $V = 8xyz$. 问题即为

求 $V = 8xyz$ 在条件 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 下的极值问题. 令

$$F(x, y, z, \lambda) = 8xyz + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right), (x > 0, y > 0, z > 0)$$

$$\begin{aligned} F'_x &= 8yz + \frac{2\lambda x}{a^2} = 0 \\ F'_y &= 8xz + \frac{2\lambda y}{b^2} = 0 \\ F'_z &= 8xy + \frac{2\lambda z}{c^2} = 0 \\ F'_\lambda &= \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0 \end{aligned}$$

解方程组 得唯一驻点 $\left(\frac{\sqrt{3}}{3}a, \frac{\sqrt{3}}{3}b, \frac{\sqrt{3}}{3}c\right)$, 即为目标函数的最大

值点. 故所求体积的最大值为 $V = \frac{8\sqrt{3}}{9}abc$.

9. 解: 令 $F(x, y, z) = 3x^2 + 2y^2 + 3z^2 - 12$, 在点 $(1, \sqrt{3}, 1)$ 处,

$$\vec{n} = (F'_x, F'_y, F'_z) \Big|_{(1, \sqrt{3}, 1)} = (6x, 4y, 6z) \Big|_{(1, \sqrt{3}, 1)} = (6, 4\sqrt{3}, 6).$$

于是所求切平面方程为

$$3(x-1) + 2\sqrt{3}(y-\sqrt{3}) + 3(z-1) = 0,$$

即

$$3x + 2\sqrt{3}y + 3z - 12 = 0.$$

所求法线方程为

$$\frac{x-1}{3} = \frac{y-\sqrt{3}}{2\sqrt{3}} = \frac{z-1}{3}.$$

10. 解: 设所求点为 $M(x_0, y_0, z_0)$, M 对应的参数值为 t_0 , 则曲面在点 M 处的法平面的法向量

为 $\vec{T} = (6t_0^2 + 2, 3t_0^2 + 2, 4)$. 由已知条件有

$$\frac{6t_0^2 + 2}{13} = \frac{3t_0^2 + 2}{7} = \frac{4}{2}$$

解得 $t_0 = \pm 2$. 从而所求点为 $(22, 13, 5)$ 和 $(-18, -11, -11)$.

曲线在点 $(22, 13, 5)$ 处的切线方程为

$$\frac{x-22}{13} = \frac{y-13}{7} = \frac{z-5}{2}.$$

曲线在点 $(-18, -11, -11)$ 处的切线方程为

$$\frac{x+18}{13} = \frac{y+11}{7} = \frac{z+11}{2}.$$

11. 解: 首先由已知条件容易得到 $f(x, y) = x^2 - y^2 + 2$. 解方程组 $\begin{cases} f'_x(x, y) = 2x = 0 \\ f'_y(x, y) = -2y = 0 \end{cases}$ 得驻点为 $(0, 0)$, 且 $f(0, 0) = 2$.

在 D 的边界 $x^2 + \frac{y^2}{9} = 1$ 上有 $x^2 = 1 - \frac{y^2}{9}$, 代入 $f(x, y)$ 得

$$G(y) = 1 - \frac{y^2}{9} - y^2 + 2 = 3 - \frac{10}{9}y^2, -3 \leq y \leq 3,$$

令 $G'(y) = -\frac{20}{9}y = 0$ 得 $y = 0$. 计算得 $G(0) = 3, G(\pm 3) = -7$. 比较可得

最小值为 $G(\pm 3) = f(0, \pm 3) = -7$, 最大值为 $G(0) = f(\pm 1, 0) = 3$.

12. 解: (1). $h(x, y)$ 在点 $M(x_0, y_0)$ 沿其梯度方向 $\text{grad } h(x_0, y_0) = (-2x_0 + y_0, -2y_0 + x_0)$ 的方向导数最大. 方向导数的最大值

$$g(x_0, y_0) = |\text{grad } h(x_0, y_0)| = \sqrt{5x_0^2 + 5y_0^2 - 8x_0y_0}.$$

(2). 问题为求函数 $g(x, y) = 5x^2 + 5y^2 - 8xy$ 在条件 $x^2 + y^2 - xy - 75 = 0$ 下的最大值. 构造拉格朗日函数:

$$F(x, y, \lambda) = 5x^2 + 5y^2 - 8xy + \lambda(x^2 + y^2 - xy - 75)$$

$$\text{解方程组} \begin{cases} F'_x = 10x - 8y + \lambda(2x - y) = 0 \\ F'_y = 10y - 8x + \lambda(2y - x) = 0 \\ F'_\lambda = x^2 + y^2 - xy - 75 = 0 \end{cases} \text{得驻点为 } (\pm 5\sqrt{3}, \pm 5\sqrt{3}) \text{ 和 } (\pm 5, \mp 5).$$

计算得 $f(\pm 5\sqrt{3}, \pm 5\sqrt{3}) = 150 < f(\pm 5, \mp 5) = 450$. 所以可以取点 $(5, -5)$ 或 $(-5, 5)$ 为登山起点.